

Customer Sales Data Cleaning & Exploratory Data Analysis

A end-to-end data cleaning and EDA project on a messy 10,400-record customer sales dataset — turning inconsistent, incomplete raw data into a clean dataset with business KPIs and visual insights.

Overview

Raw business data is rarely clean. This project takes a messy customer sales dataset (inconsistent text casing, missing values, mixed date formats, duplicate records) and runs it through a full cleaning pipeline in Python, then performs exploratory data analysis to extract business KPIs and customer behavior patterns across age groups and cities.

Problem Statement

The raw dataset had multiple data quality issues that made it unusable for direct analysis: - Missing values in Name, Email, Age, Gender, City, Purchase_Amount, and Feedback_Score - Inconsistent Gender labels (m, f, male, female in mixed case) - Inconsistent City name casing - Age stored as text with embedded non-numeric characters - 203 duplicate customer records - No pre-built KPIs or segmentation for business reporting

The goal was to clean the data, engineer relevant features, and answer: **who are the customers, where are they, and which segments drive revenue?**

Dataset

- **File:** messy_customer_sales_data.csv
- **Size:** 10,400 rows × 10 columns
- **Columns:** Customer_ID, Name, Gender, Age, City, Signup_Date, Last_Purchase_Date, Purchase_Amount, Feedback_Score, Email

Tools and Technologies

- **Language:** Python
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, re (regex)
- **Environment:** Jupyter Notebook

Methods

1. Data Cleaning - Filled missing Name and Email values with "N/A" to flag incomplete customer records rather than dropping them - Extracted numeric values from the Age column using regex, converted to integer type, and imputed missing ages with the median (to limit outlier distortion) - Standardized Gender to consistent Male / Female labels and imputed missing values with the mode - Standardized City casing and imputed missing values with the mode - Imputed missing Purchase_Amount with the median - Imputed missing Feedback_Score with the median, grouped by Gender, to preserve within-group distribution - Converted Signup_Date and Last_Purchase_Date to proper datetime format, retaining NaT where the original value was missing (no artificial dates inserted) - Removed 203 duplicate records and reset the index

2. Feature Engineering - Days_To_Last_Purchase: gap in days between signup and last purchase - Age_Group: customers segmented into Young, Adult, Senior using equal-frequency binning (pd.qcut)

3. KPI Calculation - Total, average, min, max, and median purchase amount - Unique customer count and repeat customer count - Average gap (days) between signup and last purchase - Month-over-month and year-over-year sales growth

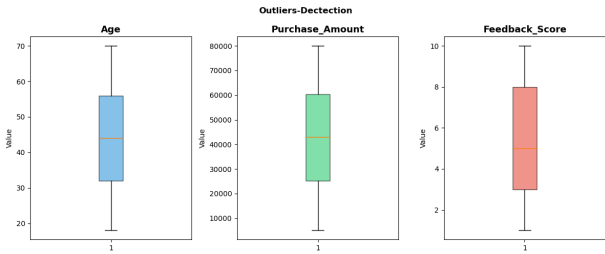
4. Exploratory Data Analysis - Outlier detection via boxplots (Age, Purchase_Amount, Feedback_Score) - Distribution analysis via histogram, boxplot, and density plots - Correlation heatmap across numeric features - Age group distribution (pie chart) - Purchase amount by age group (bar chart, line chart) - Purchase amount by city (horizontal bar chart) - City x Age group customer distribution (clustered bar chart) - Age distribution across age groups (grouped boxplot)

Key Insights

- **Customer base:** 6,390 unique customers after deduplication, with 2,727 repeat customers.
- **Revenue:** Total purchase amount $\approx$ ₹43.6 crore, average order value $\approx$ ₹42,766.
- **Purchase behavior:** Average gap between signup and last purchase is $\sim$ 1,277 days.
- **Age segmentation:** Nearly even three-way split — Young 34.5%, Senior 33.3%, Adult 32.3%.
- **Revenue by age group:** Young customers contributed the highest total purchase amount, followed by Senior, then Adult — the Adult segment is the underperformer and worth investigating.
- **Revenue by city:** Chennai leads by a wide margin ($\sim$ ₹7.08 crore), followed by Pune and Mumbai. Lucknow and Nagpur are the lowest-performing cities.
- **Correlation:** Age, Purchase_Amount, Feedback_Score, and Days_To_Last_Purchase show near-zero correlation with each other — no linear relationship exists between these variables in this dataset.
- **Outliers:** No significant outliers detected in Age, Purchase_Amount, or Feedback_Score.

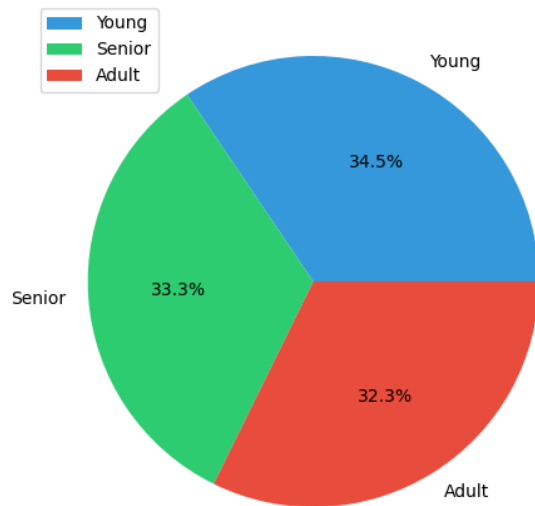
Note on data quality: The notebook's inline markdown notes stated 368 missing Name values and 437 missing Email values. Re-running the cleaning logic against the raw CSV directly shows 504 missing Name values and 655 missing Email values. If this repo is used for reporting, verify which figure is correct before citing it externally — the discrepancy likely came from how `.value_counts()` was applied in the original notebook cell rather than a change in the source data.

Dashboard / Output

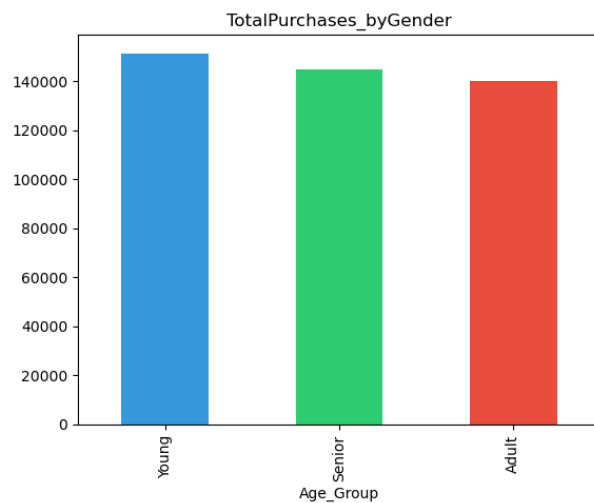
Visualization	Description
	Outlier detection across Age, Purchase_Amount, Feedback_Score

Visualization

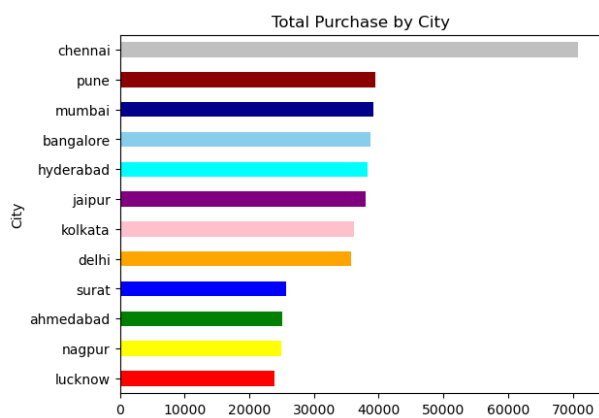
Description



Customer distribution by age group



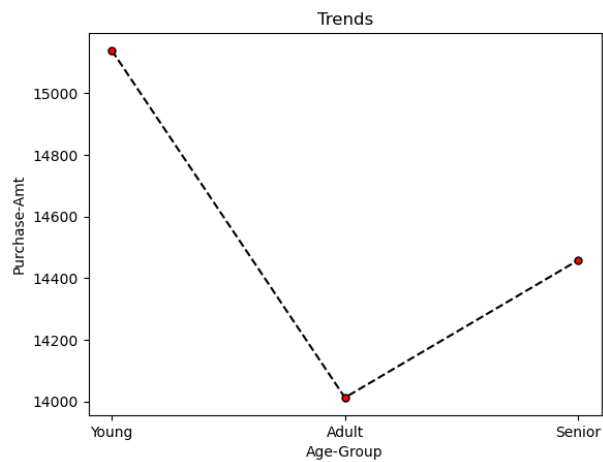
Total purchase amount by age group



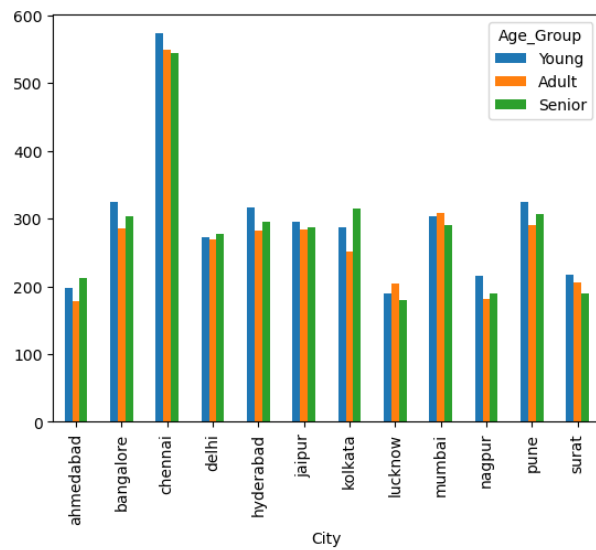
Total purchase amount by city

Visualization

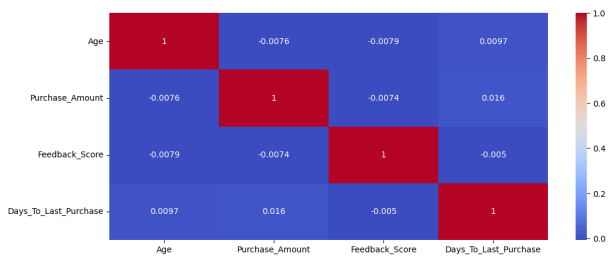
Description



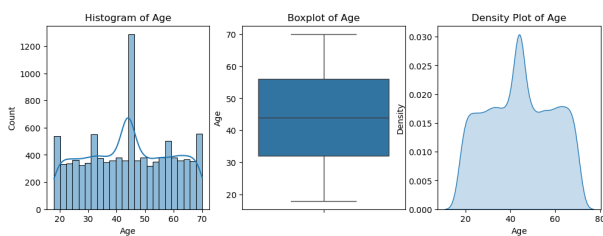
Purchase amount trend across age groups



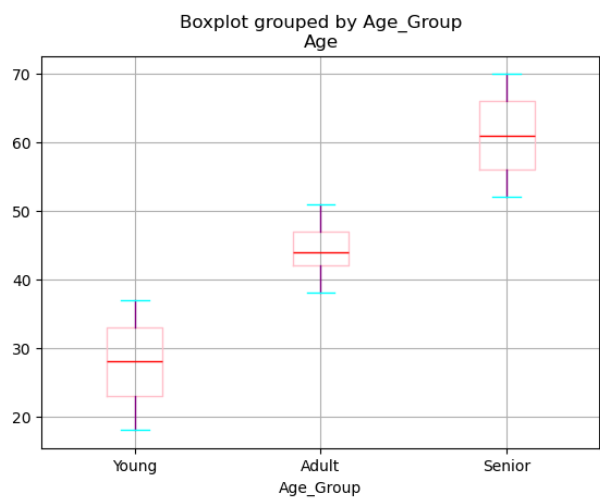
Customer distribution by city and age group



Correlation between numeric features



Histogram, boxplot, and density plot of Age

Visualization	Description
 Boxplot grouped by Age_Group Age Young Adult Senior Age_Group	Age distribution across Young, Adult, Senior groups

How to Run This Project

1. Clone the repository:

```
git clone https://github.com/NaveedSultan/<repo-name>.git
cd <repo-name>
```

2. Install dependencies:

```
pip install pandas numpy matplotlib seaborn jupyter
```

3. Launch Jupyter Notebook:

```
jupyter notebook DataCleaning.ipynb
```

4. Run all cells in order. The notebook reads `messy_customer_sales_data.csv` from the same directory, so keep the CSV alongside the notebook.

Results and Conclusion

After cleaning, the dataset went from 10,400 raw records with multiple quality issues to 10,197 clean, de-duplicated records ready for analysis. The EDA shows a business with a balanced customer base across age groups and no single segment or city dominating revenue outright — though Chennai stands out as the strongest market and the Adult age group underperforms relative to Young and Senior. The near-zero correlations across numeric features indicate that revenue and satisfaction are not being driven by age or recency of purchase in this dataset, so any targeting strategy should look beyond these variables alone (e.g. product category, purchase frequency, or channel data not present in this dataset).

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